

TREMORTRACE · PROBLEM STATEMENT

TremorTrace problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

TremorTrace problem statement

Tremor is judged from a few minutes of observation in a clinic, but it changes through the day with medication timing, fatigue, stress and activity, so the snapshot a clinician sees is often not the tremor a person lives with.

■ The problem

People with Parkinson's disease and essential tremor are typically assessed with rating scales (for example the MDS-UPDRS tremor items or TETRAS) during an appointment every few months. Those scores depend on what happens in the room that day. Patients are asked to recall how their tremor behaved between visits, and recall is poor. Medication changes are therefore judged on thin evidence.

Research has shown that a wrist-worn accelerometer and gyroscope can measure tremor frequency and amplitude continuously at home ([Mahadevan et al., npj Digital Medicine; standardized accelerometry method, PMC; rest tremor from the wrist, PMC](#)). Commercial systems exist, but they are closed, costly, or tied to one phone ecosystem. There is no open, low-cost, inspectable reference design that a researcher, clinic or patient group can build, audit and adapt.

■ Users and context

User	Need	Context
Person living with tremor	Wear it all day without thinking about it; see how tremor changed and share it	At home, daily activities, 12 to 16 h of wear
Care partner	Understand good and bad periods of the day	Home
Clinician or researcher	Objective daily tremor profile between visits, in a format they can analyze	Clinic review, research studies
Open hardware community	A reproducible design and analysis pipeline to build on	Makerspaces, university labs

■ Constraints

- Garage-buildable prototype, about \$150 USD per unit, using off-the-shelf modules and 3D-printed parts.
- Comfortable for all-day wear: light, small, skin-safe materials.
- Health data is sensitive: data stays on the device and the owner's phone by default; the owner controls any sharing.
- Research and educational use only. TremorTrace is not a medical device and must not be used to diagnose, treat or change medication.

■ Out of scope

- Diagnosis or classification of disease.
- Tremor suppression or stimulation.
- Cloud services.

■ Open questions

- Which users to involve first, and through which partner (a movement disorders clinic, a patient association, or OpenRatio's network)? Proposed, awaiting Amish.
- One wrist (most affected side) or both? Proposed: most affected side first, awaiting Amish.